

Can you create a course with curriculum materials and lab exercises incorporating the bGeigieZen (see <https://bgeigiezen.safecast.jp> for references on users manual etc.)

The bGeigieZen radiation detector by Safecast offers an excellent platform for hands-on STEM education. Below is a comprehensive course curriculum with lab exercises designed for university-level environmental science or engineering programs:

Course Title: Radiation Monitoring & Citizen Science with bGeigieZen

Module 1: Fundamentals of Radiation Detection

Learning Objectives:

- Understand basic radiation physics
- Learn device assembly and safety protocols

Lecture Topics:

- Types of ionizing radiation (α , β , γ)
- Geiger-Müller tube operation principles^[1] ^[2]
- Introduction to Safecast's open-data philosophy^[3]

Lab 1.1:

- **Kit Assembly Workshop**
 - Assemble bGeigieZen from components using official guides^[1] ^[4]
 - Test basic functionality (display, GPS acquisition)

Lab 1.2:

- **Baseline Measurements**
 - Compare background radiation:

Location	Expected Range	Measured Value
Classroom	0.05-0.2 $\mu\text{Sv/h}$	[Student Data]
Granite surfaces	0.3-0.6 $\mu\text{Sv/h}$	[Student Data]

Module 2: Technical Operation

Learning Objectives:

- Master data collection modes
- Configure advanced settings

Lecture Topics:

- GPS precision in radiation mapping^[2] ^[4]
- Real-time vs SD card logging modes^[1]

Lab 2.1:

```
# Sample code for data parsing (using Safecast API)
import requests
api_url = "https://api.safecast.org/measurements.json"
params = {"device_id": "bGeigieZen_123"}
response = requests.get(api_url, params=params).json()
print(f"Latest reading: {response[0]['value']} µSv/h")
```

Lab 2.2:

- **Mobile Survey Challenge**
 - Map campus radiation levels using:
 - 5-second mobile mode^[1]
 - Compare with fixed 5-minute interval data^[1]

Module 3: Data Analysis & Visualization

Learning Objectives:

- Process radiation datasets
- Create actionable visualizations

Lab 3.1:

- **GIS Mapping Project**
 1. Export data as CSV via Safecast API^[2]
 2. Create heatmaps using QGIS/Google Earth
 3. Identify anomalous readings requiring verification

Lab 3.2:

- **Statistical Analysis**
 - Calculate:
 - Mean radiation level:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- Standard deviation across measurement zones

Module 4: Applied Environmental Monitoring

Learning Objectives:

- Design field studies
- Interpret regulatory limits

Lab 4.1:

- **Urban-Rural Comparison**

Measurement Type	Safety Threshold
Background	<0.3 µSv/h ^[2]
Emergency	>1.0 µSv/h ^[2]

Lab 4.2:

- **Emergency Response Simulation**
 - Deploy multiple bGeigieZen units as fixed sensors ^[1]
 - Analyze real-time dashboard data ^{[1] [3]}

Module 5: Community Science Projects

Learning Topics:

- Open-source hardware development ^[5]
- Citizen science ethics

Final Project:

- **Community Radiation Atlas**
 1. Coordinate neighborhood mapping
 2. Present findings to local stakeholders
 3. Contribute data to Safecast global map ^[3]

Assessment Structure

- **40% Lab Reports**
- **30% Field Project**
- **20% Final Presentation**
- **10% GitHub Contributions** (code/docs improvements)^[5]

Resources

- [Official Documentation](#)^[6] ^[3]
- [Safecast API Guide](#)^[2]
- [GitHub Repository](#)^[5]

This curriculum combines technical training with civic engagement, leveraging the bGeigieZen's unique capabilities in real-time monitoring^[1], rugged design^[3] ^[4], and open-data integration^[2] ^[5]. The course emphasizes reproducible workflows through version-controlled code sharing^[5] and community validation of results.

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1. <https://safecast.org/safecast-introduces-our-latest-open-source-geiger-counter-the-bgeigiezen/>
2. <https://safecast.org/devices/bgeigie-zen/>
3. <https://bgeigiezen.safecast.jp>
4. <https://bgeigiezen.safecast.jp/features/>
5. <https://github.com/Safecast/bGeigieZen>
6. <https://bgeigiezen.safecast.jp/links-contact/>